

ASHWIN K K

Electronics and Communication Engineer

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SUMMARY

Electronics and Communication Engineer with hands-on knowledge in the areas of embedded systems, IoT, and wireless communication technologies. Proficient in Embedded C, ESP32, Arduino, PCB design, soldering and electronic circuit testing. Familiar with communication protocols. Seeking opportunities to develop innovative and reliable engineering solutions.

SKILLS

- **Programming Languages:** Embedded C, SQL
 - **Communication Protocols:** UART, SPI, I2C
 - **Tools:** kiCad, MPLAB, MS office
 - **Soft skills:** Communication, Teamwork, Observation, Creativity, Adaptability
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EDUCATION

B. Tech in Electronics and Communication Engineering	2022 – 2026
College of Engineering Thalassery	
Diploma in Electronics Engineering	2020 - 2022
Government Polytechnic College Mattanur	

PROJECTS

LoRa-based V2X Communication System	2025 - 2026
• Developed a V2X communication prototype using ESP32, Arduino Nano, and LoRa (SX1278). • Implemented V2V, V2I, and V2P communication for traffic and safety applications. • Generated collision-warning, traffic-signal, and pedestrian alert notifications.	
Blackbox System for Emergency Response	2024 - 2025
• Developed a vehicle SOS prototype system for accident detection and emergency response. • Sent emergency alerts with GPS location, date, and timestamp information. • Stored accident data in memory and evaluated sensor response accuracy.	
Vision-Based Intelligent Automation System	2021 - 2022
• Developed a vision-based automation system using the YOLO algorithm for face detection. • Automatically controlled lights and fans based on human presence detection. • Improved energy efficiency and monitored the number of people inside halls.	

CERTIFICATES

Technical Support Fundamentals [Google]

- Learned the basics of technical support, troubleshooting, and solving common IT-related problems.
- Improved problem-solving skills through practical technical support examples and activities.
- Gained confidence in identifying technical issues and providing proper support solutions.

Introduction to IoT and Embedded Systems [University of California, Irvine]

- Learned the basics of IoT devices, embedded systems, and communication technologies..
- Gained knowledge of processors, sensors, memory, and communication protocols.
- Understood how connected devices communicate and exchange data in IoT applications.

Embedded system Design and Programming [Invade Embedded Technologies]

- Completed an internship program focused on embedded systems and microcontroller programming.
- Worked on PIC microcontroller programming, sensor interfacing, and PCB design activities.
- Developed practical experience in embedded hardware and software development.